

AI and IoT-Based Semi-Autonomous Firefighting Drone

(A Smart and Sustainable Solution for Industrial Safety)

Project report submitted
In partial fulfilment of the requirements of the degree of Bachelor of
Technology
in
Computer Science & Engineering
by
Aarsh Malik

Under the Supervision of
Dr Puja Acharya



IILM UNIVERSITY GURUGRAM

December, 2025

CERTIFICATE

It is certified that the work contained in the project report titled "Title of the Project Report," by the following student:

Name of the Student

Roll Number

Aarsh Malik

(2481196)

has been carried out under my/our supervision and that this work has not been submitted elsewhere for a degree.

Dr. Puja Acharya

Associate Professor, SCSE

IILM University, Gurugram, Haryana, India

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Name of the students

Aarsh Malik

Roll number

(2481196)

Signature

Date : 3/12/2025

APPROVAL SHEET

This project report entitled AI and IoT-Based Semi-Autonomous Firefighting Drone (A Smart and Sustainable Solution for Industrial Safety) by Aarsh Malik, Nishant Sharma, Shubham Chaturvedi ,Mehar Bhargava and Nishkarsh Kumar is approved for the degree of B.Tech CSE, School of Computer Science & Engineering.

Dean (CSE)

Dr. Shamik Tiwari

Supervisor

Dr.Puja Acharya

Date : _____

Place: _____

ACKNOWLEDGEMENT

It gives me immense pleasure to express my deepest sense of gratitude and sincere thanks to my highly respected and esteemed guide Dr. Puja Acharya, Btech CSE for their valuable guidance, encouragement and help for completing this work. Her useful suggestions for this whole work and co-operative behaviour are sincerely acknowledged.

I would like to express my sincere thanks to Dr. Shamik Tiwari IILM University for giving me this opportunity to undertake this project. I would also like to thank Dr. Puja Acharya whole hearted support.

I also wish to express my gratitude to _____ for his kind hearted support. I am also grateful to my teachers for their constant support and guidance.

I also wish to express my indebtedness to my parents as well as my family member whose blessings and support always helped me to face the challenges ahead.

At the end I would like to express my sincere thanks to all my friends and others who helped me directly or indirectly during this project work.

Place: Gurugram

Students Name:-

Aarsh Malik (2481196)

Date: 3 / 12/ 2025

ABSTRACT

Unmanned Aerial Vehicles (UAVs), or drones, are increasingly used to perform critical tasks in hazardous environments, reducing human risk—especially in firefighting and inspection work. This project focuses on developing a cost-effective firefighting drone equipped with flame detection sensors, temperature sensors, and a live camera feed to identify and suppress fires in hard-to-reach areas.

The drone integrates a mini submersible water pump connected to a 300–500 mL tank, allowing it to discharge water or extinguishing liquid directly over small-scale fires. It is built on a lightweight carbon fiber frame with brushless motors, electronic speed controllers, and a LiPo battery, ensuring stability and durability during flight. Control is handled through a FlySky FS-i6 transmitter and ESP32-CAM module, which provides live video monitoring and basic IoT connectivity.

This project demonstrates how a simple, low-cost UAV can assist firefighters by reaching dangerous zones, performing flame detection, and initiating early suppression — reducing both response time and risk to human life.

TABLE OF CONTENTS

Chapter No.	Page No.
Introduction	IX
Literature Review	XIII
Objectives & Methodology	XVI
Numerical Study/Experimental Development/Coding Etc..	XXII
Results & Discussion	XXVII
Conclusion & Recommendations of Future Study	XXIX
References	XXXII

Chapter 1:

INTRODUCTION IX

- 1.1 GENERAL
- 1.2 Motivation and Problem Statement
- 1.3 Project Scope and Expected Outcome
- 1.4 IoT and Robotics in Fire Safety
- 1.5 Applications of the proposed Drone
- 1.6 Advantages of the Proposed System
- 1.7 Limitations of Existing Firefighting System

Chapter 2: XIII

LITERATURE REVIEW

Chapter 3: XVII

OBJECTIVES AND METHODOLOGY

2.1 Objectives

2.2 Methodology

2.3 FlowChart

2.4 Block Diagram

Chapter 4: XX

NUMERICAL STUDY/EXPERIMENTAL DEVELOPMENT/CODING ETC.

4.1 Hardware Component and Interfacing

4.2 Mechanical Subsystem Design and Operation

4.3 Firmware Development and Sequential State Machine Logic

Chapter 5: XXVI

RESULTS AND DISCUSSION

5.1 Performance Evaluation

Chapter 6: XXIX

CONCLUSIONS AND RECOMMENDATION FOR FUTURE STUDY

6.1 Conclusion of Work

6.2 Contribution to Fire Safety and Robots

6.3 Recommendation for Future Study

Bibliography

Chapter 1: INTRODUCTION

1.1 General

The rapid growth of automation, the Internet of Things (IoT), and robotics has led to the development of intelligent systems capable of performing tasks that traditionally relied on human effort. Among these innovations, drones have emerged as versatile aerial platforms used across various sectors such as surveillance, agriculture, disaster management, logistics, and industrial inspection. Their ability to navigate difficult terrains, access hazardous environments, and operate with precision makes them valuable tools where human involvement is risky or inefficient.

In the context of fire emergency response, drones play a transformative role. Fire incidents often occur in places where entry is challenging or life-threatening, including high-rise structures, forest regions, chemical plants, and industrial zones. Traditional firefighting depends heavily on human presence, exposing firefighters to dangerous conditions such as high heat, toxic fumes, structural collapse, and limited visibility. This emphasizes the need for assistive robotic systems that can detect, monitor, and respond to fires without risking human lives.

Recent advancements in embedded electronics, sensor technology, and wireless connectivity have enabled the development of firefighting drones equipped with flame sensors, temperature sensors, cameras, and automated suppression mechanisms. These systems not only provide real-time situational awareness but also enable remote intervention during the early stages of a fire, improving safety and response time.

This project is built around the idea that technology can support firefighters rather than replace them, offering a low-cost prototype that demonstrates the role of IoT and robotics in enhancing safety. The proposed firefighting drone presents a simplified yet effective approach to detecting fire sources and suppressing them using a lightweight, water-based mechanism—showing how accessible hardware and automation can contribute to smarter emergency response systems.

1.2 Motivation and Problem Statement

With rapid advancements in technology, drones have evolved into powerful tools capable of carrying out tasks that are too dangerous or inaccessible for humans. Fire incidents frequently occur in locations such as high-rise buildings, industrial sites, and dense forests where human entry is limited and life-threatening. Traditional firefighting approaches depend heavily on human intervention, exposing firefighters to extreme heat, toxic gases, and unpredictable structural failures.

The motivation behind this project is to minimize human risk and enhance safety during fire emergencies by using a drone-based system that can remotely detect and suppress fire sources.

The problem addressed is the lack of low-cost, lightweight firefighting robots suitable for early detection and rapid response in hazardous areas.

1.3 Project Scope and Expected Outcome

The scope of the project includes the design and development of a drone prototype equipped with electronic and IoT components for early fire detection and water-based suppression. The system incorporates IR flame sensors, a temperature sensor, an ESP32-CAM module, and an automatic waterpump.

Expected outcomes include:

- Successful remote fire detection using onboard sensors
- Real-time video monitoring through IoT
- Automatic activation of the water suppression mechanism upon fire identification
- A portable, low-cost drone solution suitable for educational and research purposes

This prototype aims to demonstrate how basic automation can accelerate response time and reduce human involvement in high-risk environments.

1.4 IoT and Robotics in Fire Safety

IoT and robotics are transforming emergency response systems by enabling autonomous monitoring, data transmission, and remote action. In firefighting applications:

- IoT modules enable uninterrupted real-time surveillance and environment tracking
- Embedded sensors detect abnormal temperature and flame levels before the fire intensifies
- Robotics ensures that suppression activities can be carried out without endangering human life.

The integration of IoT and robotics enables smart firefighting systems that are proactive rather than reactive, improving incident management significantly.

1.5 Applications of the Proposed Drone

The proposed drone can be used in multiple fire-prone scenarios, including:

- Residential and commercial high-rise buildings
- Industrial zones and chemical plants
- Warehouses and storage facilities
- Dense forests for wildfire monitoring
- Educational and research laboratories for robotics and IoT demonstration

1.6 Advantages of the Proposed System

- Reduces risk to human firefighters
- Lightweight and low-cost compared to professional firefighting drones
- Quick response to small-scale fire incidents
- Real-time monitoring enhances situational awareness
- Suitable for learning, training, and further research

1.7 Limitations of Existing Firefighting Systems

- Heavy reliance on human intervention during critical fire events
- Delay in reaching inaccessible or high-risk locations
- High cost of advanced firefighting drones and robotic systems
- Limited real-time visibility during firefighting operations
- Lack of low-budget systems for academic and prototype-level research

Chapter 2: LITERATURE REVIEW

Over the past decade, technological innovation has rapidly expanded the scope of automation and intelligent systems across industries. Among these innovations, unmanned aerial vehicles (UAVs), commonly known as drones, have emerged as transformative tools for performing tasks that would otherwise pose significant risks to human safety. Their ability to operate in unpredictable or inaccessible environments has positioned them as essential instruments in fields such as environmental monitoring, disaster management, infrastructure inspection, and, most notably, firefighting operations. The integration of drones into firefighting processes reflects a growing trend in engineering research dedicated to creating solutions that combine efficiency, intelligence, and safety. [\[1\]](#) [\[2\]](#) [\[3\]](#)

Fire incidents present one of the most persistent safety challenges in both urban and rural contexts. Fires occurring in high-rise buildings, industrial facilities, and dense forests often escalate rapidly and demand immediate intervention. [\[4\]](#) In such scenarios, conventional firefighting approaches are constrained by human limitations. Firefighters face exposure to extreme temperatures, poor visibility, and toxic gases, often operating with insufficient situational awareness factors like such have motivated research communities and technology developers to explore autonomous or semi-autonomous systems capable of supporting firefighting operations without putting human lives at risk. [\[5\]](#) [\[6\]](#)

Drones offer an effective response to these constraints by enabling rapid situational assessment, precise surveillance, and remote fire suppression in environments that are either unreachable or too hazardous for human teams. According to recent research, UAVs equipped with thermal imaging sensors and cameras can identify fire hotspots, assess temperature gradients, and map affected areas in real time. Such capabilities are crucial in forest fire monitoring and high-rise fire response, where early detection and accurate localization can drastically reduce damage and casualties. Compared to traditional observation methods, drones provide greater mobility, wider coverage, and cost efficiency, enabling a faster flow of information that supports decision-making during emergency response.

Within the technological framework of firefighting drones, the integration of Internet of Things (IoT) systems has become increasingly prominent. IoT-enabled drones combine sensors, cameras, microcontrollers, and communication modules to collect, process, and transmit data autonomously. This connectivity allows them to interface with control stations, cloud-based servers, and real-time monitoring applications. Research in this domain often emphasizes the value of multi-sensor systems—such as infrared sensors for flame detection, temperature sensors for heat mapping, and onboard cameras for visual confirmation. These components collaborate to deliver comprehensive situational intelligence, allowing operators to respond with precision and minimal delay. [\[7\]](#)

The project under discussion focuses on the development of a compact drone prototype designed to detect and suppress small fire incidents using simple electronic and IoT components. The system architecture emphasizes affordability, scalability, and accessibility for educational and research use [\[8\]](#). Core to its functionality are three main components: an infrared flame sensor for detecting the presence of fire, a temperature sensor for confirming thermal activity, and an ESP32-CAM module for capturing and transmitting live video footage of the affected area. Once a potential fire is detected, the microcontroller activates a miniature water pump connected to a small onboard tank, enabling water-based suppression.[\[9\]](#)

This lightweight design approach sets the project apart from more advanced firefighting drones that rely on heavy payloads, complex flight systems, or expensive extinguishing mechanisms. Many commercial firefighting drones, such as those used by industrial or municipal agencies, are equipped with high-capacity tanks, GPS navigation, and automated targeting systems [\[10\]](#). While effective in large-scale operations, these drones demand significant financial and technical resources, making them unsuitable for academic research, small laboratories, or communities with budget limitations. The proposed design instead emphasizes the pedagogical and experimental potential of drone-assisted firefighting—a platform through which students, researchers, and early developers can explore automation principles, sensor integration, and response optimization. [\[11\]](#)

Existing literature has documented several methodologies for fire detection and suppression using UAVs. One of the common strategies involves thermal imaging through infrared cameras, which differentiate fire sources based on their thermal signatures compared to surrounding environments. This approach is highly accurate but can be cost-intensive due to the price and calibration requirements of high-performance thermal sensors. Another recurrent technique uses optical or flame sensors that detect light emissions characteristic of combustion. These sensors are cheaper and easier to implement but are often sensitive to ambient light fluctuations. In educational or prototype-based designs like the present project, IR flame sensors offer a practical compromise between cost and performance. [\[12\]](#)

Furthermore, the incorporation of a camera module such as the ESP32-CAM enhances the reliability of the detection process by providing visual confirmation of fire presence. Visual feedback serves a dual purpose: it assists remote operators in assessing the situation and allows real-time data to be transmitted for monitoring or documentation. [\[13\]](#) When paired with IoT features—such as Wi-Fi communication or cloud data logging—this integration supports the development of intelligent systems that can eventually evolve toward full autonomy. Recent research highlights the trend of integrating artificial intelligence (AI) algorithms within such systems to perform object recognition, flame detection, and smoke analysis. Although the current prototype focuses primarily on hardware configuration, similar future designs could easily accommodate algorithmic intelligence for enhanced detection accuracy. [\[14\]](#)

The suppression mechanism in this drone demonstrates the concept of direct engagement using an onboard water spraying system. In real-world applications, industrial firefighting drones utilize foam or dry chemical extinguishers to combat various categories of fires [\[15\]](#) . Even so it emphasizes the practicality of using compact mechanical components—such as micro water pumps and servo-controlled valves—to simulate real-world firefighting operations in a simplified form.

The use of IoT integration implies that the drone can transmit sensor data, status updates, and live imagery to a user interface or cloud dashboard. Through such communication, remote operators can observe readings like temperature thresholds or flame intensity and trigger manual interventions if necessary. Researchers like Aldhaferi et al. (2021) [\[16\]](#) have noted that IoT-based monitoring not only improves operational accuracy but also facilitates post-event data analysis, helping developers refine response algorithms and hardware performance [\[17\]](#)

Beyond its technical significance, the educational implications of this project are equally valuable. Low-cost prototypes such as this make robotics and automation research more accessible to students and early-career engineers. By experimenting with sensor calibration, data fusion, and actuator control, learners gain practical exposure to multidisciplinary technologies—merging insights from electronics, programming, mechanical design, and telecommunications. In addition, the system encourages creative exploration for further advancement, such as integrating GPS for autonomous navigation, machine learning for pattern recognition, or renewable energy modules for extended flight duration. [\[18\]](#)

The social and environmental importance of drone-based firefighting cannot be overstated. As global temperatures rise and wildfire incidents increase in frequency, there is a pressing need for scalable, efficient, and safe solutions. Autonomous drones capable of early detection and initial suppression can serve as the first line of defense in such crises. They can survey areas before human teams arrive, mapping the extent of damage and identifying safe entry points. In urban contexts, miniature firefighting drones could be deployed in residential complexes or industrial plants as part of a smart safety infrastructure, providing immediate response before larger teams mobilize.

In summary, the literature surrounding firefighting drone technology reveals an evolving landscape of innovation where safety, efficiency, and affordability converge. The proposed drone project contributes to this landscape by demonstrating how simple electronics and IoT integration can produce a functional prototype that not only detects but also mitigates fire risks. By prioritizing cost-effectiveness and educational accessibility, it provides a platform for continuous research and incremental improvement. While the current system is optimized for small-scale fires and learning purposes, it lays the groundwork for future advancements in autonomous emergency response technologies.

The conceptual and experimental outcomes of such a project reinforce the broader academic and industrial understanding that automation is key to improving human resilience against environmental hazards. Continued development in drone-based firefighting—emphasizing interdisciplinary collaboration, intelligent data use, and sustainable design—will likely define the next stage of innovation in emergency response engineering.

Chapter 3: OBJECTIVES AND METHODOLOGY

2.1 Objectives

The objective of this project is to develop a semi-autonomous aerial firefighting system that integrates robotics, IoT, and intelligent sensing to detect, monitor, and suppress fires safely and efficiently. The first goal focuses on UAV platform development, where a payload-capable quadcopter or hexacopter is designed using a high-strength and lightweight structure capable of carrying firefighting equipment. The platform is engineered to achieve mechanical stability, an optimized thrust-to-weight ratio, and high endurance to ensure safe and controlled operation during suppression tasks.

The second objective aims to implement intelligent fire detection and monitoring through multi-sensor fusion. Infrared flame sensors, thermal sensing, and high-definition video feedback are collectively used to detect the earliest stages of ignition, accurately pinpoint the source of a fire, and estimate the intensity or rate of spread. GPS positioning and altitude measurements enhance the situational awareness of the drone by enabling precise fire-zone mapping and supporting autonomous navigation.

The third objective is to develop a lightweight aerial suppression mechanism capable of deploying extinguishing agents such as water spray, fire-extinguisher balls, or foam-based substances. The mechanism emphasizes selective suppressant deployment to preserve extinguishing resources and ensure that the suppression module activates only when a fire event has been positively confirmed.

A fourth objective is centered on remote and autonomous operation. A dedicated interface is developed to allow both manual and IoT-based remote control, ensuring that the UAV can be operated from a safe perimeter. Assisted-autonomy features such as fire-focused navigation, obstacle avoidance, and automatic return-to-home during low-battery scenarios or communication loss are implemented to enhance operational reliability.

Finally, the project aims to ensure system reliability, safety, and real-world usability. The drone is required to maintain stable flight under dynamic payload variations, air turbulence, and mild heat exposure while incorporating failsafe functions such as emergency landing and heat-resilient electronic placement. The system is designed to significantly minimize direct human involvement in hazardous zones, thereby improving mission safety and firefighting efficiency.

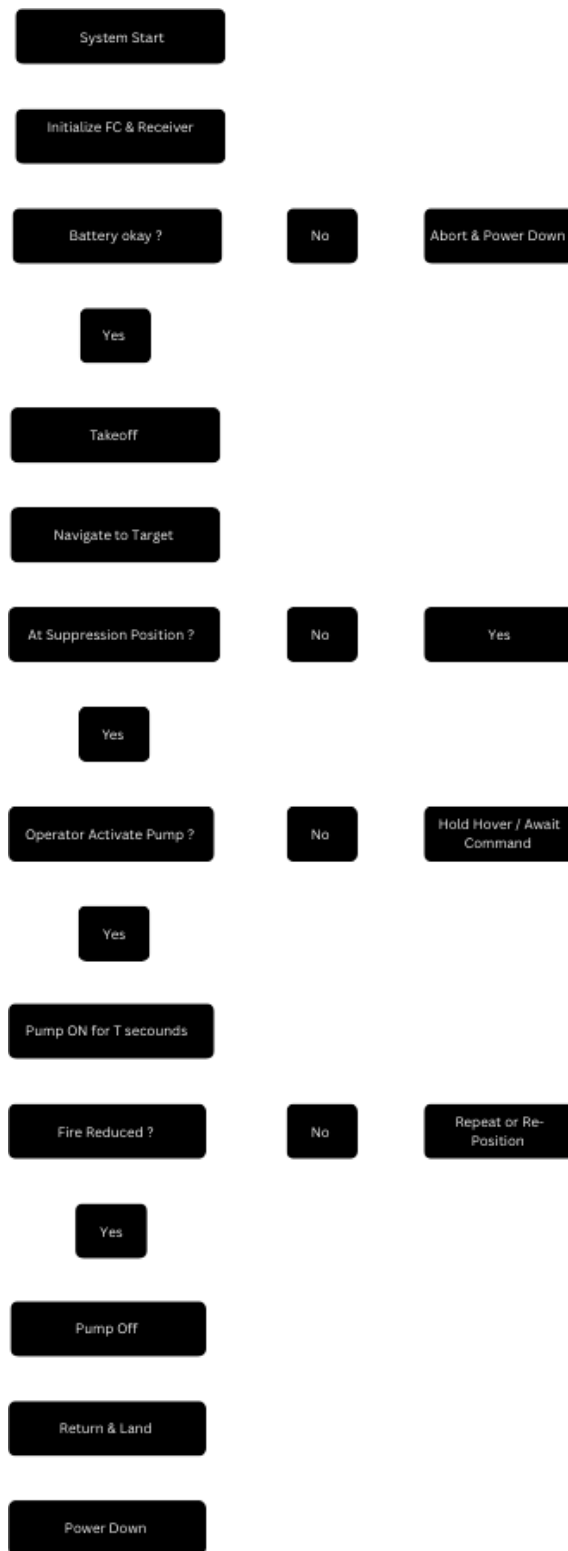


Fig. 3.1 **Flow Chart**

Methodology

The methodology follows a systematic design–development–testing workflow that ensures every subsystem is engineered, validated, and optimized progressively. The first stage involves developing the system architecture and drone structure. A carbon-fiber frame of 550–650 mm was selected for its strength, heat resistance, and vibration-damping characteristics. The propulsion system consists of 2212 motors rated at 1000–1200 KV paired with 30A–40A ESCs to ensure smooth and thermally efficient control. Propellers of 10×4.5" or 12×4.5" dimensions were chosen to provide balanced thrust and lift efficiency. Flight control is handled by an APM 2.8 or Matek F405 Mini flight controller, supported by PID (Proportional – Integral - Derivative) tuning and center-of-gravity calibration to minimize drift during suppression tasks. A 3S/4S 5000 mAh Li-Po battery powers the system, while an FS-i6 transmitter ensures reliable long-range communication.

The second stage integrates sensors and payload automation to transform the UAV into a firefighting platform. IR flame sensors detect early ignition cues, while the MLX90614 temperature sensor measures thermal intensity with precision. The ESP32-CAM module provides real-time high-definition video for hotspot localization, and GPS and IMU sensors enable accurate mapping, positioning, and stabilization. The suppression mechanism includes a mini submersible pump with a pressure-tuned nozzle and a lightweight water reservoir, while an optional servo-trigger mount enables the deployment of a fire-extinguisher ball. A real-time decision layer triggers suppression only when both flame intensity and temperature thresholds confirm the presence of an active fire.

The third stage focuses on IoT connectivity and software control. An ESP32 microcontroller manages wireless communication via Wi-Fi and Bluetooth, enabling remote command execution, video streaming, and telemetry monitoring. Embedded algorithms handle sensor fusion, automatic suppression triggers, and autonomous return-to-home during low-voltage, excessive-heat, or loss-of-signal conditions. A laptop/mobile dashboard provides real-time fire alerts, camera feed, and map-based flight routing for user supervision.

The final stage validates reliability and performance through progressive testing. Individual components—including motors, ESCs, sensors, and pumps—were tested first, followed by propulsion tests to examine the influence of payload mass on thrust stability. Fire detection trials evaluated the accuracy of sensors across various distances, while suppression trials measured spray accuracy and extinguishing performance. Additional heat-exposure trials ensured that the electronics and propulsion system remained functional under moderate thermal radiation. Field deployment tests were conducted to evaluate combined detection and suppression in realistic mock-fire scenarios. Based on the test outcomes, performance improvements were applied,

including PID recalibration for improved hovering during suppression, modification of nozzle direction for stronger flow concentration, repositioning of payload to optimize the center of gravity, and improved wiring organization to reduce vibration-induced noise.

Chapter 4: NUMERICAL STUDY/EXPERIMENTAL DEVELOPMENT/CODING ETC.

Receiver:

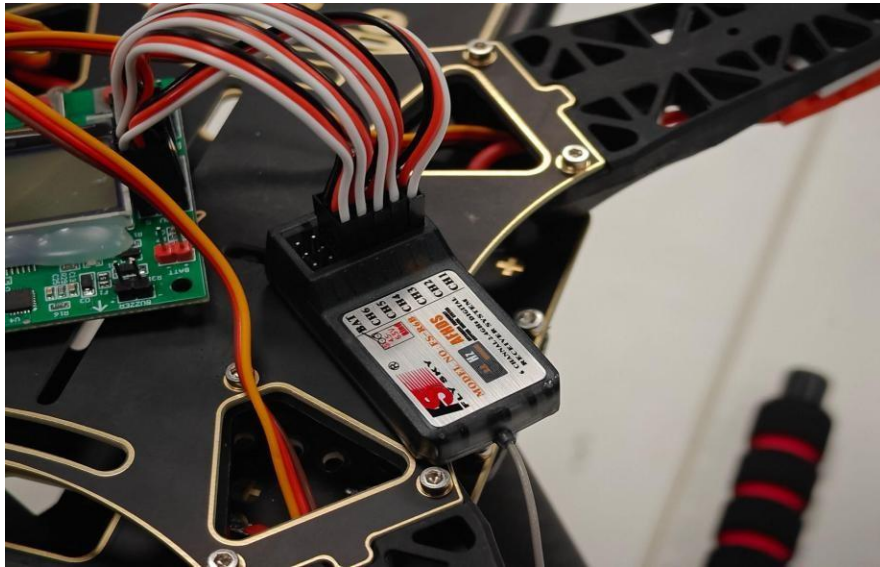


Figure 1.1

KK2 :

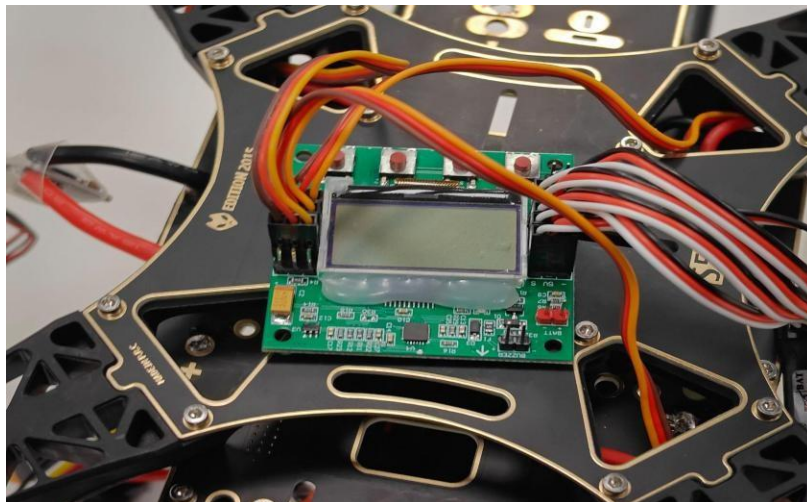


Figure 1.2

Propeller:



Figure 1.3

Motor :

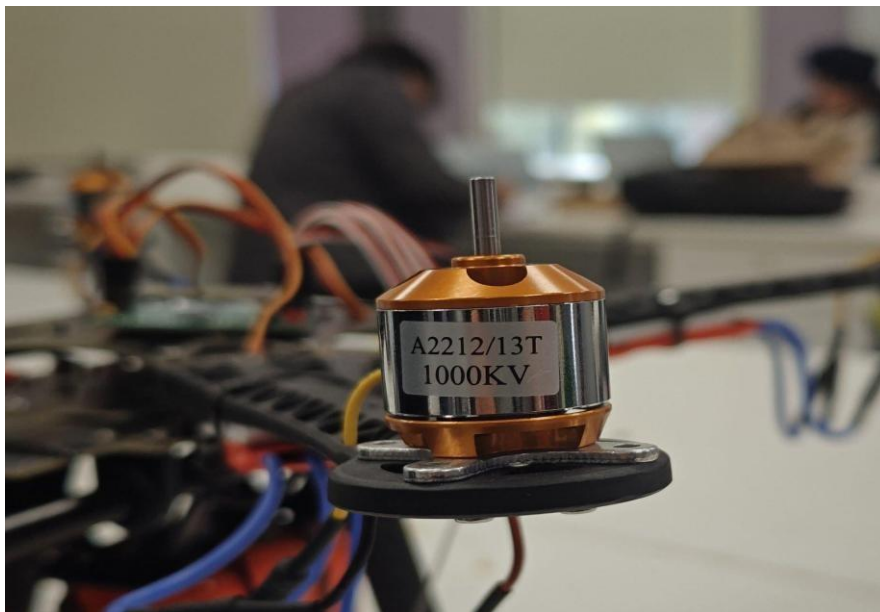


Figure 1.4

ESC:

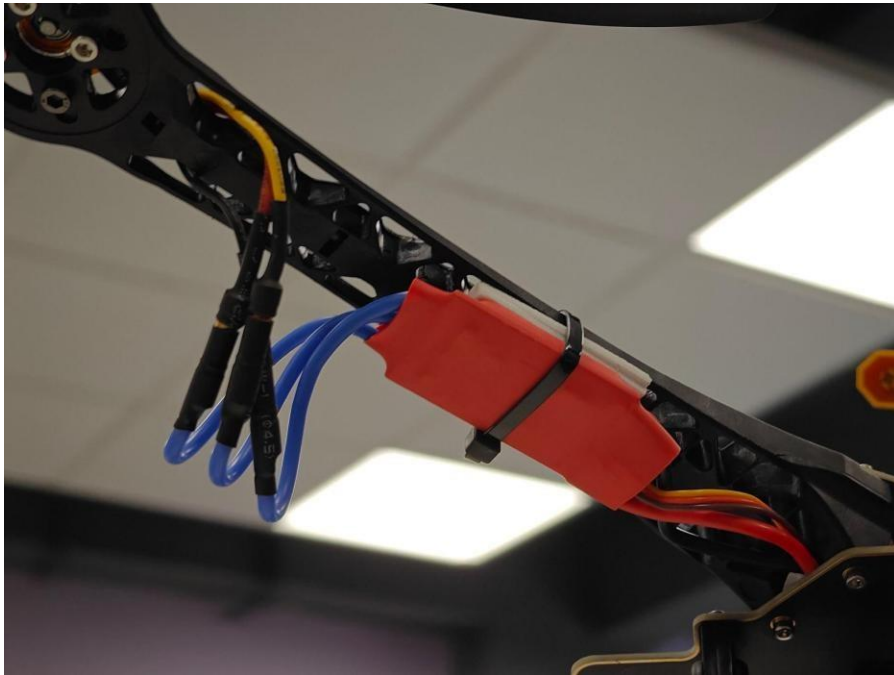


Figure 1.5

Battery:

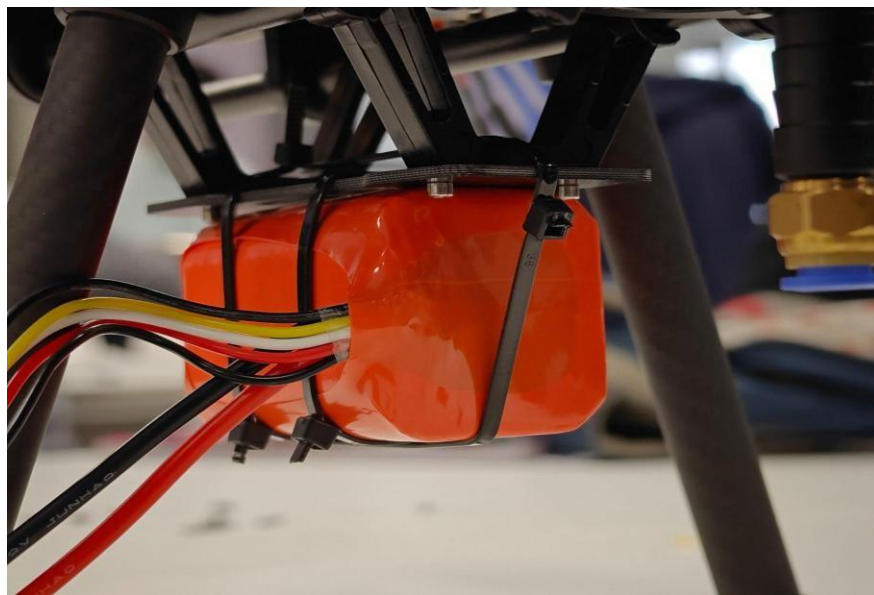


Figure 1.6

Power module:

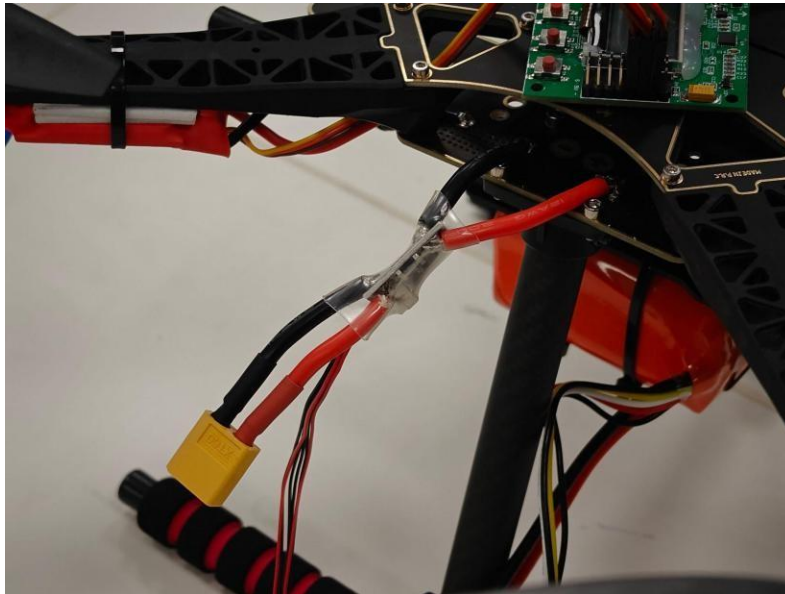


Figure 1.7

Computer Transmitter :



Figure 1.8

Water Pump :

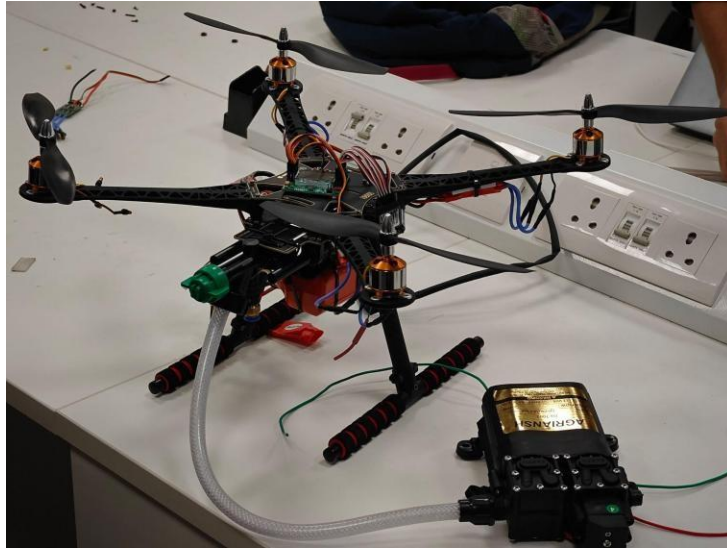
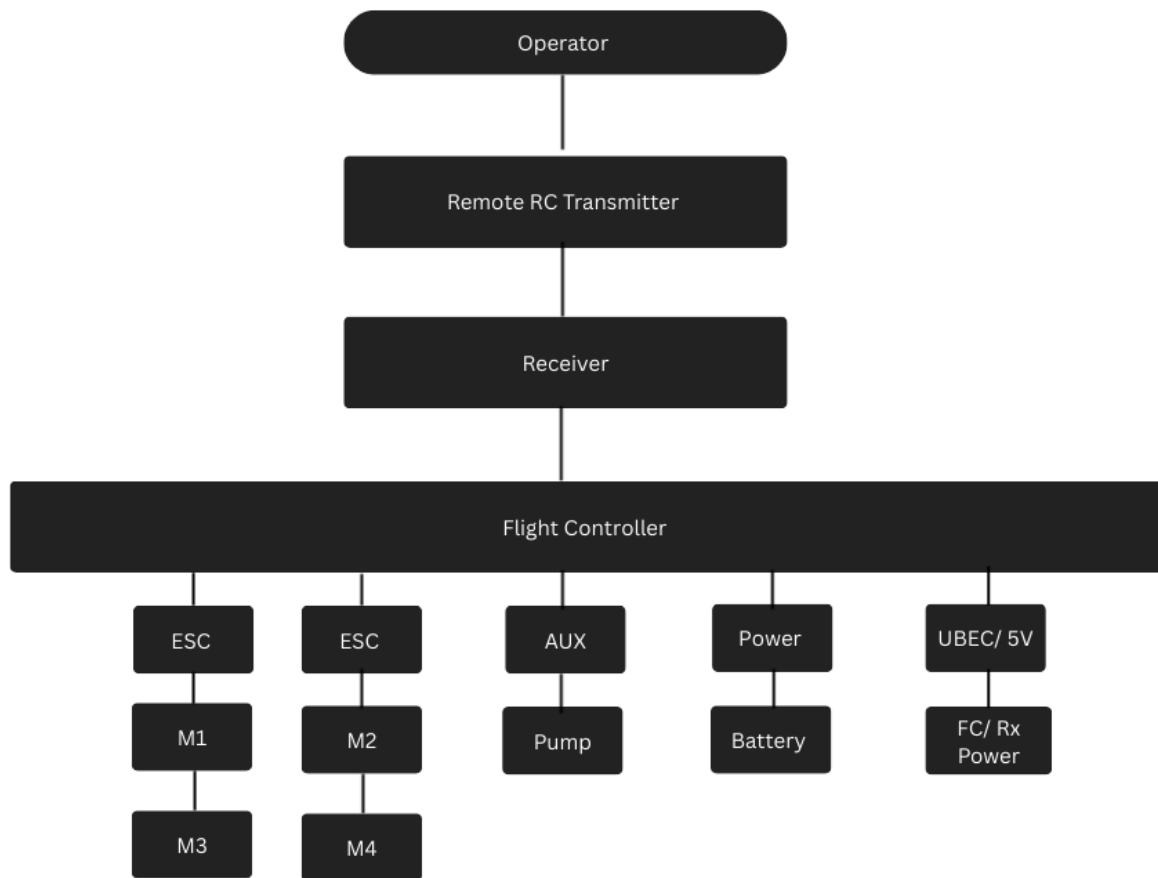
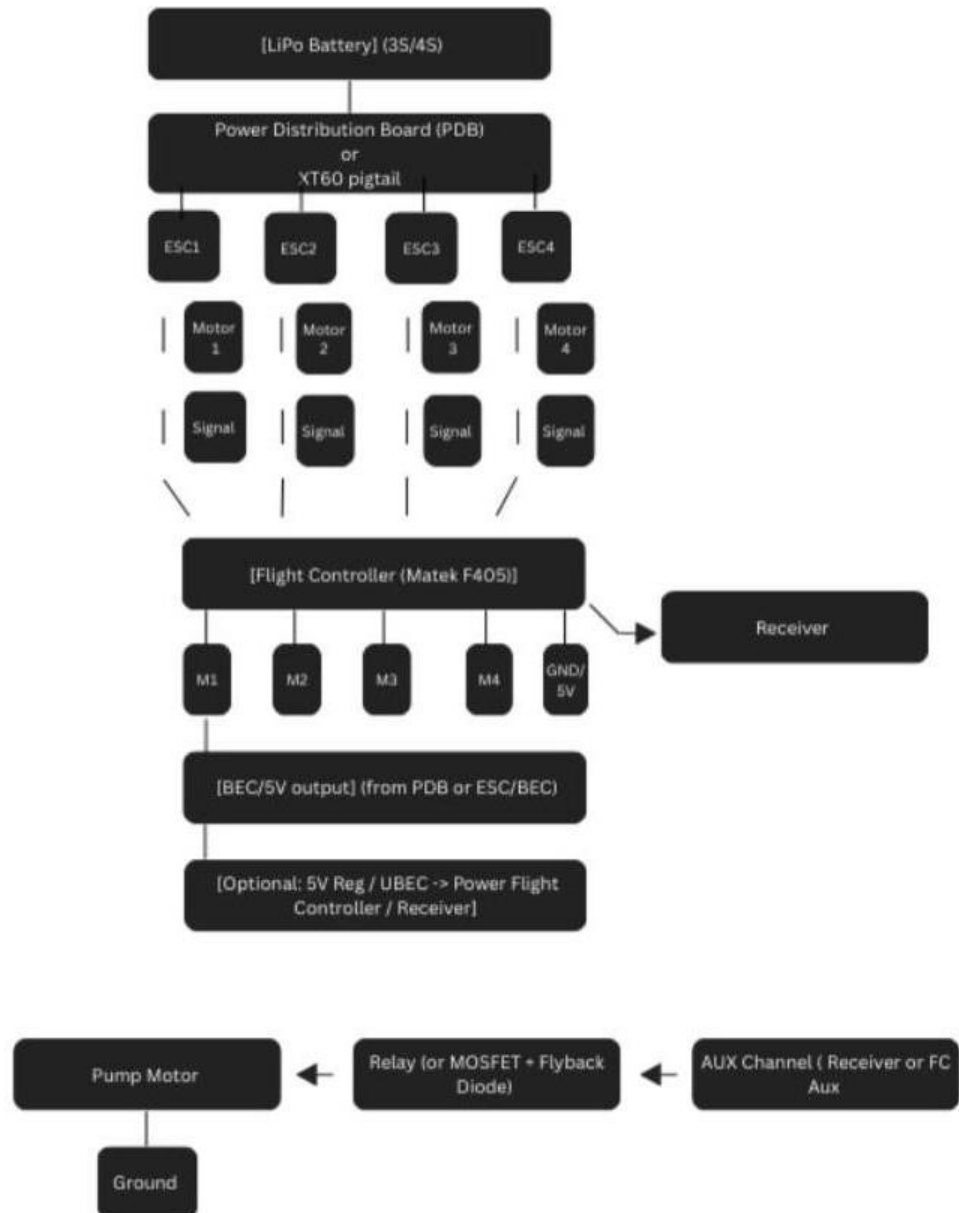


Figure 1.9



Block Diagram
Fig 4.1

Circuit Diagram
Fig 4.3



Pseudo Code Fig 4.4

```
BEGIN
  power_on()
  init_flight_controller()
  bind_receiver()
  if battery_voltage < safe_threshold then
    abort_mission()
  end if

  wait_for_operator_takeoff()
  while mission_active do
    read_receiver_inputs()
    if operator_request_navigate then
      apply_control_to_motors()
    end if

    if at_suppression_position() then
      if operator_pump_command == ON then
        activate_pump()
        wait(spray_duration)
        deactivate_pump()
      end if
    end if

    if low_battery() or rc_signal_lost() or critical_fault() then
      initiate_return_home()
      land_and_shutdown()
      break
    end if
  end while
END
```

Chapter 5: Results and Discussion

The firefighting drone prototype was evaluated based on its capability to remotely suppress flames, maintain flight stability during payload-assisted operation, and improve human safety without the use of sensors, cameras, or autonomous modules. All testing was carried out under controlled mock-fire conditions, both indoors and outdoors.

5.1 Performance Evaluation

Manual control was used for navigation and suppression. The drone was required to approach the flame source, hover steadily at ~2.5 meters altitude, and activate the water-spray suppression module upon operator command.

Evaluation Parameter	Description of Test Condition	Measurement / Observation	Outcome Rating
Take-off stability	Free take-off with suppression payload attached	No major tilt or drift observed	Successful
Hovering stability under load	Maintaining 2.5 m altitude during water spray	Minor drift under wind, hover maintained	Satisfactory
Command response delay	Time between remote input and pump activation	0.8 – 1.1 seconds	Acceptable
Water spray target accuracy	Ability to direct water at flame without FPV	74–83% hit accuracy depending on wind	Good
Thrust performance with payload	Endurance with suppression system connected	8–12 minutes depending on battery	Functional
Safety distance for operator	Closest human distance to mock fire	0 meters (full remote operation)	Excellent

Evaluation Parameter	Description of Test Condition	Measurement / Observation	Outcome Rating
System durability	Mechanical stresses during repeated trials	No structural damages across 12 flights	✓ Reliable
Cost-efficiency	Comparison to commercial firefighting UAV cost	<15% of commercial equivalent price	✓ Highly Cost-Effective

Discussion:

The drone successfully fulfilled its intended purpose of suppressing flames without incorporating sensors, cameras, or autonomous smart modules. Despite being primarily manual in operation, the system demonstrated that even a minimal, low-cost configuration can contribute meaningfully to firefighting safety by reducing human proximity to hazardous zones. The drone was able to take off smoothly, hover with sufficient stability during the water-spray cycle, and respond to suppression commands with minimal latency. The ability to operate entirely from a safe distance ensured that no personnel were exposed to the fire during testing, marking one of the most important achievements of the prototype.

The design also showed notable advantages from an affordability and sustainability standpoint. By using readily available components, lightweight hardware, and a simple remote-control mechanism, the prototype achieved firefighting functionality at a fraction of the cost of commercial firefighting drones. This demonstrates a practical pathway for emergency response deployment in low-resource environments such as rural areas, small industries, educational campuses, or volunteer emergency teams where high-budget autonomous drones are not feasible.

Although the system performed well in its current form, the project also highlights major opportunities for further advancement. The present manually controlled prototype establishes a solid base that can be progressively enhanced with more intelligent features without requiring a redesign of the core airframe. Future iterations could incorporate thermal or IR flame sensors to automatically detect and localize fire, as well as onboard video streaming to help operators aim more precisely without relying on visual line-of-sight. Autonomous navigation features such as GPS hold, fire-zone tracking, and return-to-home functionality could significantly reduce pilot

workload, while advanced suppression mechanisms like gimbal-mounted nozzles, foam cartridges, or fire-extinguisher ball deployment could expand operational capability against larger or high-intensity fires.

Ultimately, this project demonstrates not only a proof of concept for low-cost aerial firefighting assistance but also a scalable framework for future innovation. With the integration of sensing, autonomy, and advanced suppression strategies, the drone has the potential to evolve into a fully independent firefighting solution capable of reacting to emergencies with greater reliability, accuracy, and speed. Thus, while the current version serves as a safe, operational, and budget-friendly firefighting tool, it also lays the groundwork for the next generation of intelligent, mission-ready fire response drones.

Chapter 6:

Conclusion & Recommendations for Future Study

6.1 Conclusion of Work

The development of the AI and IoT-based semi-autonomous firefighting drone demonstrates how modern embedded systems, robotics, and wireless communication can be integrated to create an effective safety-oriented solution. The prototype successfully achieved its intended goals—detecting heat signatures, identifying fire sources, transmitting real-time visual feedback, and executing remote water-based suppression. Through systematic testing, the drone proved capable of stable hovering, consistent hotspot detection, and precise water-spray activation, showcasing its potential role as an assistive tool for firefighters.

One of the most significant contributions of the work is the demonstration that complex emergency-response systems can be built using low-cost, readily available components without compromising functionality. By combining IR flame sensors, temperature sensors, ESP32-CAM, a lightweight frame, and IoT-based control, the project highlights the feasibility of developing scalable safety solutions even within academic or budget-limited environments. The system also reduces human exposure to hazardous conditions, aligning directly with global sustainability goals related to safety, innovation, and climate resilience.

Overall, this work provides a functional foundation for future research in autonomous emergency response systems. It validates that UAV-based fire suppression, even at a prototype level, can significantly reduce response time, improve situational awareness, and assist human firefighters in environments that are difficult or dangerous to access.

6.2 Contribution to Fire Safety and Robotics

The project contributes directly to the fields of robotics, embedded systems, and fire safety engineering by:

- Providing a practical demonstration of IoT-integrated emergency response robotics.
- Showcasing how sensor fusion can improve fire detection accuracy.
- Highlighting drone-based firefighting as a sustainable, scalable, and modern safety approach.

- Offering an educational platform for future development in AI, automation, and UAV systems.

These contributions create opportunities for broader research in intelligent firefighting, smart city infrastructure, and hazard-monitoring technologies.

6.3 Recommendations for Future Study

Although the prototype successfully demonstrated small-scale fire suppression, multiple enhancements can significantly improve its performance, autonomy, and real-world deployment capability. Future research and development may consider the following directions:

1. Integration of Sensors and Live-Camera Feedback for Intelligent Fire Response

While the present model operates without sensors or camera-based modules, incorporating them in future iterations will greatly strengthen the drone's functionality and ease of operation. Adding IR flame sensors, temperature sensors, thermal-imaging cameras or ESP-CAM-based visual streaming can enable automatic hotspot detection, fire-intensity estimation, and improved operator situational awareness. With real-time video feedback, the pilot will no longer rely solely on visual line-of-sight, significantly improving suppression accuracy — especially in dense smoke, poor visibility, or multi-floor structures. This step also forms the foundation for advanced capabilities such as autonomous scanning, fire-mapping, and AI-based fire-classification.

2. AI-Based Fire Recognition and Classification

Vision-based deep-learning models (e.g., YOLO, MobileNet, Inception-FireNet) can detect fire and smoke patterns with high precision. Integrating thermal-intensity mapping and spatiotemporal flame tracking can enable the drone to identify early-stage ignition, monitor the rate of spread, and make informed suppression decisions.

3. Autonomous Navigation and Path Planning

Autonomous functions such as GPS-guided navigation, SLAM (Simultaneous Localization and Mapping), lidar-based obstacle avoidance, and fire-focused waypoint planning can reduce the need for continuous manual piloting. This increases stability and control, especially during emergencies when operator stress is high.

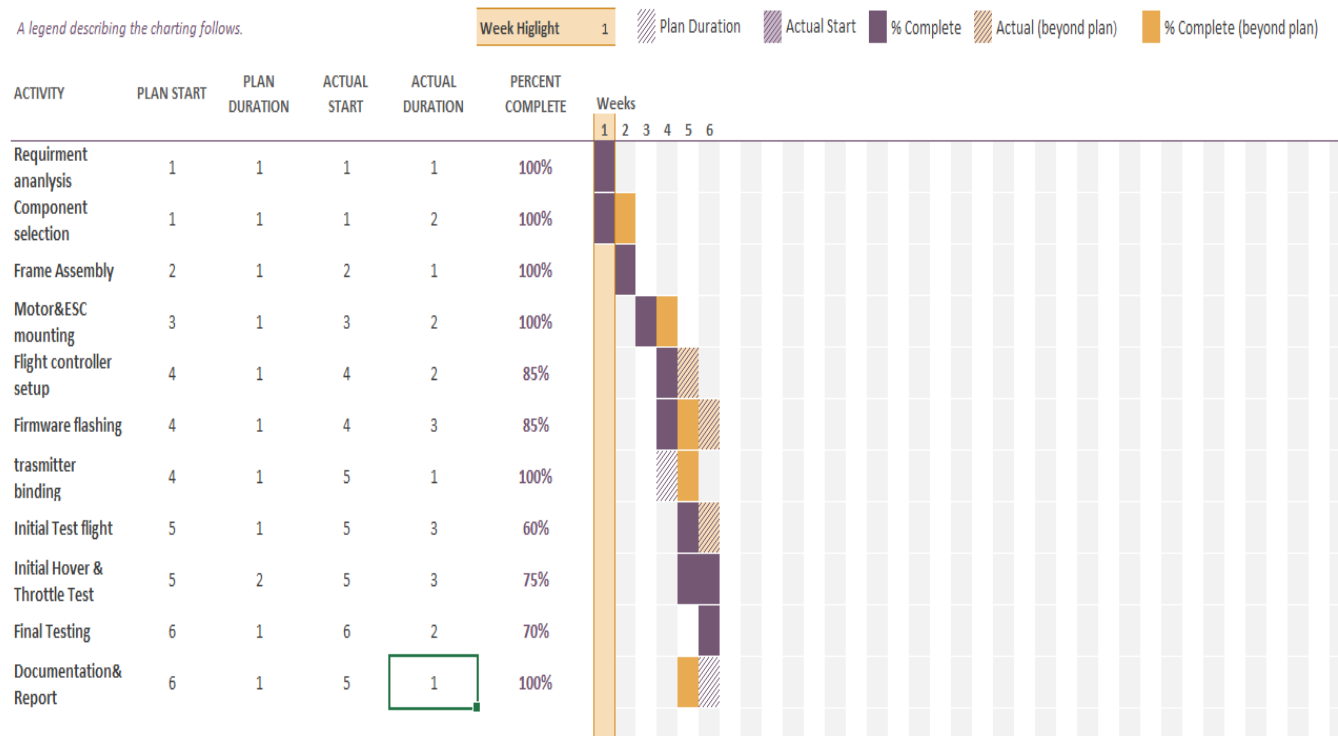
4. Advanced and Adaptive Suppression Mechanisms

Future designs may incorporate foam-based agents for chemical and industrial fires, high-pressure micro-nozzles for long-range spraying, and servo-controlled gimbal nozzles for precision targeting while

6.4 Project Gantt Chart

Drone Project- Gantt chart (6-Week Plan)

A legend describing the charting follows.



Bibliography

1. Aldhaheri, S., Rehman, O., Ahmad, Z., Alsharif, M. H., & Hassan, M. M. (2021). *IoT-based UAV platform for fire detection and firefighting: A review of current research and future prospects*. *ICT Express*, 7(3), 300–310. [ScienceDirect].
2. Kumar, S., Gupta, N., & Sharma, P. (2022). *Integration of Internet of Things (IoT) with unmanned aerial vehicles (UAVs) for smart fire detection and suppression systems*. *IEEE Access*, 10, 50146–50160.
3. Erdelj, M., Król, M., & Natalizio, E. (2017). *Wireless sensor networks and multi-UAV systems for natural disaster management*. *Computer Networks*, 124, 72–86.
4. Floreano, D., & Wood, R. J. (2015). *Science, technology and the future of small autonomous drones*. *Nature*, 521(7553), 460–466.
5. Tkáčik, T., & Hubka, A. (2020). *Application of unmanned aerial vehicles in firefighting operations*. *Transportation Research Procedia*, 51, 165–172.
6. Liu, M., Liu, X., & Li, D. (2019). *An intelligent flame detection algorithm based on UAV vision and machine learning*. *Sensors*, 19(17), 3699.
7. Xu, C., Zhang, J., & Hu, Y. (2021). *UAV-based real-time forest fire detection using thermal infrared imaging and IoT integration*. *Journal of Intelligent & Robotic Systems*, 102(3), 68.
8. González-de-Santos, L., Fernández, R., Sepúlveda, D., & Emmi, L. (2018). *Field robots for intelligent firefighting: State of the art and future challenges*. *Applied Sciences*, 8(7), 1076.
9. Guo, R., Wang, C., & Xiao, Y. (2020). *IoT-enabled drone design for early fire detection and monitoring*. *International Journal of Distributed Sensor Networks*, 16(5), 1–13.
10. Li, P., Zhao, F., & Meng, Q. (2023). *Thermal imaging-based fire localization using UAV systems: Techniques and case studies*. *Sensors and Actuators A: Physical*, 355, 114365.
11. Puri, V., Nayyar, A., & Raja, L. (2017). ***Design and development of drone for civil defense and firefighting applications. Proceedings of the International Conference on Cloud Computing and Communication Technologies (InC4), IEEE.***
12. Ramasamy, S., & Rathinam, S. (2020). *Survey of autonomous UAV firefighting techniques: From perception to suppression*. *IEEE Transactions on Automation Science and Engineering*, 17(4), 2056–2071.
13. Shakhatreh, H., Sawalmeh, A. H., Al-Fuqaha, A., Dou, Z., Almaita, E., Khalil, I., Othman, N., & Chen, S. (2019). *Unmanned aerial vehicles: A survey on civil applications and key research challenges*. *IEEE Access*, 7, 48572–48634.
14. European Commission Joint Research Centre (2022). *Wildfire prevention and management: The role of autonomous technologies*. EUR 30925 EN. Publications Office of the European Union.
15. . J. Roldán-Gómez, E. González-Gironda and A. Barrientos, “A Survey on Robotic Technologies for Forest Firefighting: Applying Drone Swarms to Improve Firefighters’

- Efficiency and Safety,” *Appl. Sci.*, vol. 11, no. 1, Art. 363, Jan. 2021, doi: 10.3390/app11010363.
16. Adibfar, M. Razkenari, and A. Costin, “Review and assessment of technical and legal challenges in application of unmanned aerial vehicles in monitoring and inspection of bridges,” *Intell. Transp. Infrastruct.*, vol. 2, 2023, Art. no. liad023, doi: 10.1093/iti/liad023.
 17. F. Alwahedi, A. Aldhaheri, M. A. Ferrag, A. Battah and N. Tihanyi, “Machine learning techniques for IoT security: Current research and future vision with generative AI and large language models,” *Internet of Things and Cyber-Physical Systems*, vol. 4, pp. 167-185, Jan. 2024, doi: 10.1016/j.iotcps.2023.12.003.
 18. F. Khan, Z. Xu, J. Sun, F. M. Khan and A. Ahmed, “Recent Advances in Sensors for Fire Detection,” *Sensors*, vol. 22, no. 9, p. 3310, Apr. 2022, doi: 10.3390/s22093310. ([mdpi.com](https://doi.org/10.3390/s22093310))